

**Prepare:** FNRP Ch. 3 (multigroup equations); review Lesson 12

**Primary objective.** Construct multigroup cross sections and solve neutron balances for an infinite, homogeneous system.

## 1 Replace assumed flux ratios

Lesson 12 supplied normalized shapes and *assumed* the flux in each range. Now determine those fluxes from coupled balances. Number groups from **high to low energy**:

$$E_0 > E_1 > \dots > E_G, \quad \phi_g = \int_{E_g}^{E_{g-1}} \phi(E) dE.$$

The complete domain has  $E_0 = \infty$  and  $E_G = 0$ . For our three-group examples, use the following finite bounds (eV):

$$(E_0, E_1, E_2, E_3) = (10^7, 10^5, 1, 10^{-3}).$$

Thus  $(\phi_1, \phi_2, \phi_3) = (\phi_F, \phi_I, \phi_T)$ . Finite cutoffs require attention to transfers outside the retained range.

## 2 Preserve each group reaction rate

For  $\phi_g > 0$ , define

**Group averages**

$$\Sigma_{x,g} = \frac{\int_{E_g}^{E_{g-1}} \Sigma_x(E) \phi(E) dE}{\phi_g},$$

$$(\nu \Sigma_f)_g = \frac{\int_{E_g}^{E_{g-1}} \nu(E) \Sigma_f(E) \phi(E) dE}{\phi_g}.$$

Use a normalized within-group shape to prepare these averages; then solve for  $\phi_g$  (Lesson 12). The group sources are

$$\chi_g = \int_{E_g}^{E_{g-1}} \chi(E) dE, \quad s_g = \int_{E_g}^{E_{g-1}} s_{\text{ext}}'''(E) dE.$$

## 3 Keep the scattering direction straight

**Destination  $g \leftarrow$  incident group  $g'$**

$$\Sigma_{s,g \leftarrow g'} = \frac{\int_{E_g}^{E_{g'-1}} \phi(E') \int_{E_g}^{E_{g-1}} \Sigma_s(E' \rightarrow E) dE dE'}{\phi_{g'}}.$$

The denominator is the **incident-group flux**. Multiplying by  $\phi_{g'}$  recovers the rate of transfer into  $g$ .

### Checkpoint

With this numbering, downscatter means  $g$  \_\_\_\_\_  $g'$ . In a scattering matrix whose rows are destinations, are the forced zero (upscatter) entries above or below the diagonal?

## 4 One balance per destination group

Integrate the spectrum equation over  $E_g \leq E \leq E_{g-1}$ :

**Infinite-medium multigroup balance**

$$\Sigma_{t,g} \phi_g = \sum_{g'=1}^G \Sigma_{s,g \leftarrow g'} \phi_{g'} + \chi_g \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} + s_g, \quad g = 1, \dots, G.$$

The fission term sums production over *all incident groups*;  $\chi_g$  then distributes the newborn neutrons to destination  $g$ . Move self-scattering to the left by defining **removal**:

$$\Sigma_{r,g} = \Sigma_{t,g} - \Sigma_{s,g \leftarrow g}.$$

With number conservation and complete energy coverage,

$$\Sigma_{r,g} = \Sigma_{a,g} + \sum_{h \neq g} \Sigma_{s,h \leftarrow g}.$$

Removal includes absorption *and* scattering out of a group; it is not generally the same as absorption.

## 5 Three-group water tank

Approximate the  $^{252}\text{Cf}$  source as  $(s_1, s_2, s_3) = (10, 0, 0)$  in  $\text{cm}^{-3} \text{s}^{-1}$ . Ignore multiplication, leakage, intergroup upscatter, and transfers outside the finite energy range.

**Solve from fast to thermal**

Write each group balance and isolate its flux:

$$\phi_1 = \frac{\Sigma_{s,3 \leftarrow 1} \phi_1 + \Sigma_{s,3 \leftarrow 2} \phi_2}{\Sigma_{r,3}}.$$

Unlike the Lesson 12 example, the flux ratios are now outputs.

**Check the complete balance**

Add the three equations. Under the stated assumptions, show that

$$\sum_{g=1}^3 \Sigma_{a,g} \phi_g = 10 \text{ cm}^{-3} \text{s}^{-1}.$$

Why should removal cross sections *not* replace absorption cross sections in this check?

## 6 Use the scattering kinematics

For elastic scattering isotropic in the CM system from a stationary target of mass ratio  $A$ , let  $\alpha = [(A - 1)/(A + 1)]^2$ . Then

$$p(E' \rightarrow E) = \begin{cases} 1/[(1 - \alpha)E'], & \alpha E' \leq E \leq E', \\ 0, & \text{otherwise.} \end{cases}$$

This kernel cannot upscatter; it is not a thermal free-gas kernel. The allowed outgoing interval and destination group overlap over

$$\Delta E_g(E') = \left[ \min(E_{g-1}, E') - \max(E_g, \alpha E') \right]_+, \quad [z]_+ = \max(z, 0).$$

Thus the probability of landing in group  $g$  is

$$P_g(E') = \frac{\Delta E_g(E')}{(1 - \alpha)E'}.$$

### Incident-flux-weighted transfer

$$\sigma_{s,g \leftarrow g'} = \frac{\int_{E_{g'}}^{E_{g'-1}} \sigma_s(E') P_g(E') \phi(E') dE'}{\int_{E_{g'}}^{E_{g'-1}} \phi(E') dE'}.$$

For a mixture, compute each nuclide's transfer with its own  $\alpha_i$ , then form  $\Sigma_{s,g \leftarrow g'} = \sum_i N_i \sigma_{s,g \leftarrow g'}^i$ .

### An empty overlap is zero

Why would directly integrating from the larger lower bound to the smaller upper bound give an unphysical result?

## 7 Hydrogen: scatter out of group 2

Use the same three-group bounds,  $^1\text{H}$  ( $\alpha = 0$ ),  $\sigma_s(E) = \sigma_0 = 20$  b, and  $\phi(E') \propto 1/E'$  within incident group 2. Write

$$L = \ln(E_1/E_2) = \ln(10^5).$$

### Set up the two nonzero integrals

$$\sigma_{s,2 \leftarrow 2} = \frac{\sigma_0}{L} \int_{E_2}^{E_1} \frac{E' - E_2}{(E')^2} dE',$$

$$\sigma_{s,3 \leftarrow 2} = \frac{\sigma_0}{L} \int_{E_2}^{E_1} \frac{E_2 - E_3}{(E')^2} dE'.$$

Explain the numerators using the allowed outgoing energy intervals.

## 8 Evaluate the scattering column

The group-to-group cross sections are

$$\begin{aligned} \sigma_{s,1 \leftarrow 2} &= 0, \\ \sigma_{s,2 \leftarrow 2} &= \sigma_0 \frac{L + E_2/E_1 - 1}{L}, \\ \sigma_{s,3 \leftarrow 2} &= \sigma_0 \frac{(E_2 - E_3)(1/E_2 - 1/E_1)}{L}. \end{aligned}$$

### Compute the values

| Destination            | $\sigma_{s,g \leftarrow 2}$ (b) |
|------------------------|---------------------------------|
| $g = 1$ (fast)         | 0                               |
| $g = 2$ (intermediate) | _____                           |
| $g = 3$ (thermal)      | _____                           |

Most collisions stay within the very broad intermediate group; about 8.68% enter the retained thermal interval. These are conditional collision fractions, *not* ratios of steady-state group fluxes.

### Where did the missing cross section go?

With  $E_3 = 10^{-3}$  eV, some hydrogen collisions produce  $0 < E < E_3$ . Their contribution is

$$\sigma_{s,<E_3 \leftarrow 2} = \sigma_0 \frac{E_3(1/E_2 - 1/E_1)}{L} = \text{_____ b.}$$

Hence

$$\sum_{g=1}^3 \sigma_{s,g \leftarrow 2} = \sigma_0 - \sigma_{s,<E_3 \leftarrow 2} \approx 19.99826 \text{ b.}$$

How would the column sum change if the last group extended to  $E = 0$ ?

## 9 Check the model, not just the algebra

For a complete set of outgoing energies,

$$\sum_g \Sigma_{s,g \leftarrow g'} = \Sigma_{s,g'}, \quad R_x = \sum_g \Sigma_{x,g} \phi_g.$$

For truncated groups, keep any out-of-range transfer as an additional loss, or state the approximation used to close the range. Do not silently claim exact conservation after discarding that transfer.

### Takeaway

Multigroup balance determines the *amount* of flux in each group. The within-group spectra used to prepare cross sections remain modeling inputs. Fission couples the groups again; Lesson 14 uses that coupling to construct the  $k$ -eigenvalue problem.